

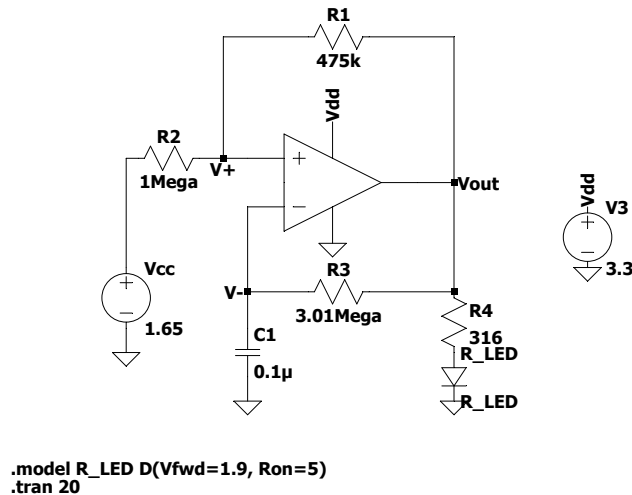
ENGR 3430: Blinker

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The blinker was first designed and simulated in LTspice before the circuit was transferred into KiCad. The simulation was used to select the resistor and capacitor values, verify the Schmitt-trigger thresholds and oscillator period, and confirm that the output would drive the LED as expected.

1 Thresholds

The oscillator uses an inverting Schmitt trigger with positive feedback. Two $10\text{ k}\Omega$ resistors divide the 3.3 V supply to create a nominal 1.65 V reference. A $475\text{ k}\Omega$ feedback resistor and a $1\text{ M}\Omega$ resistor connected to the reference set the switching thresholds at approximately 0.531 V and 2.769 V . Because these values are nearly symmetric around 1.65 V , the timing capacitor spends approximately equal amounts of time charging and discharging, producing a duty cycle near 50% .



--- C:\Users\ihong\OneDrive - Olin College of Engineering\Documents\LTspice\Draft5.asc ---

Figure 1: LTspice schematic used to select and verify the oscillator component values.

2 Oscillator Period and Worst-Case Analysis

The blink period is set primarily by the $3.01\text{ M}\Omega$ timing resistor and $0.1\text{ }\mu\text{F}$ timing capacitor:

$$\tau = RC = (3.01\text{ M}\Omega)(0.1\text{ }\mu\text{F}) = 0.301\text{ s.}$$

Using the nominal thresholds, the total period is approximately

$$T \approx 2RC \ln \left(\frac{2.769}{0.531} \right) = 0.994 \text{ s.}$$

The LTspice transient simulation shows the timing voltage repeatedly moving between the two thresholds while the op-amp output switches between its high and low states.

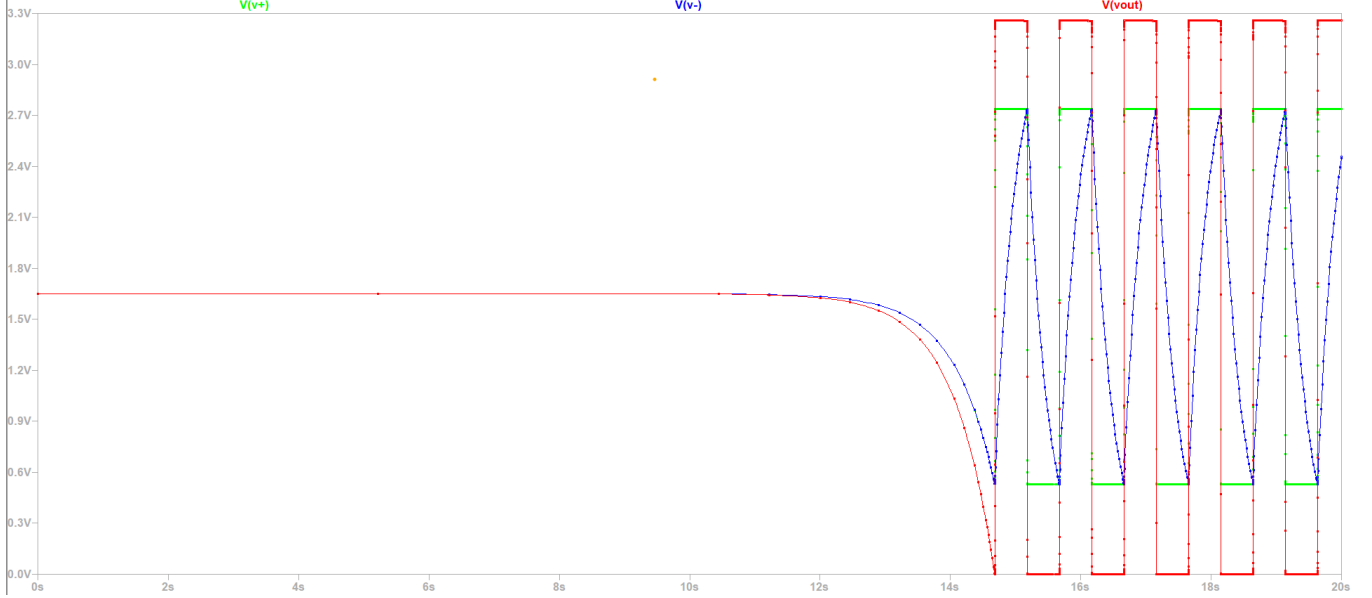


Figure 2: LTspice transient simulation showing the two switching thresholds, timing voltage, and output voltage.

A worst-case estimate was calculated using per-unit sensitivity. With the reference centered at $V_{DD}/2$, the period can be written in terms of the $1 \text{ M}\Omega$ reference resistor R_{ref} , the $475 \text{ k}\Omega$ feedback resistor R_{fb} , the timing resistor R , and timing capacitor C :

$$T = 2RC \ln \left(1 + 2 \frac{R_{\text{ref}}}{R_{\text{fb}}} \right).$$

The period is directly proportional to R and C , so their per-unit sensitivities are both 1. Their $\pm 1\%$ and $\pm 5\%$ tolerances therefore contribute approximately 1% and 5% period variation. The per-unit sensitivities to R_{ref} and R_{fb} are approximately $+0.490$ and -0.490 , so their $\pm 1\%$ tolerances each contribute about 0.49%. The equal $10 \text{ k}\Omega$ divider places the reference at $V_{DD}/2$, where small reference shifts have approximately zero first-order effect on the total period.

Adding the magnitudes of the first-order contributions gives a conservative period variation of

$$\frac{\Delta T}{T} \approx 5\% + 1\% + 0.49\% + 0.49\% = 6.98\%.$$

The resulting estimated range is

$$T_{\text{min}} \approx 0.924 \text{ s}, \quad T_{\text{max}} \approx 1.063 \text{ s.}$$

3 Choosing Op Amps and Regulators

The MCP6021 op amp was selected because it operates from a 3.3 V single-ended supply and provides rail-to-rail input and output operation. This allows the Schmitt trigger to use most of the available 0 V to 3.3 V range without the input or output limitations of other available op amps. The MCP1702-3302 regulator converts the 5 V USB VBUS supply to the 3.3 V rail used by the oscillator.

4 Adding Capacitors

Additional capacitors were added according to the component datasheets. The $0.1\ \mu\text{F}$ op-amp bypass capacitor provides a local source of current during fast switching and reduces supply noise. The $1\ \mu\text{F}$ capacitors at the regulator input and output provide bulk capacitance and regulator stability. A separate $0.1\ \mu\text{F}$ capacitor bypasses the $1.65\ \text{V}$ reference so that switching noise does not significantly shift the Schmitt-trigger thresholds. These capacitors serve different purposes from the timing capacitor, whose value directly controls the blink period.

5 PCB Layout

The datasheets also guided component placement. The USB connector was placed at the board edge, with the regulator and its input and output capacitors nearby. The op amp, bypass capacitor, feedback resistors, and timing components were grouped closely together. In particular, the bypass capacitor was placed within the recommended distance from the op-amp supply pin. Short connections reduce trace resistance and inductance and limit noise coupling into the high-impedance timing and threshold nodes.

6 Design Files

<https://drive.google.com/drive/folders/1Tt9hSKRzCtY8MVjkrJQ5e5fssUZTR6hn?usp=sharing>